WORKED SOLUTION

STEP II 2012, Question 7

Pure • Vectors and Circles

STEP II 2012, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

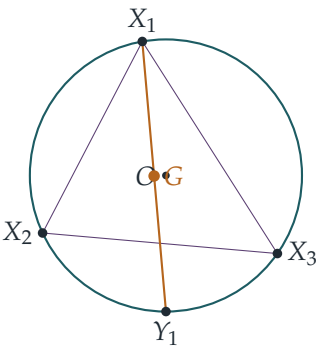
step.maths.org/questions/12-s2-q7

The key idea

The point Y_1 lies on the line through X_1 and G, so its position vector is a simple combination of $\mathbf{x}_1$ and $\mathbf{g}$, with one unknown scalar. The only other fact we know about Y_1 is that it is on the unit circle, so $|\overrightarrow{OY_1}|^2 = 1$ gives an equation for λ_1. It is a quadratic, but one root is obvious in advance: the line meets the circle at X_1 itself, so that root can be factored out.

Solution

Write $\mathbf{g} = \frac{1}{3}(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3)$ and $\lambda = \lambda_1$. Since each $\mathbf{x}_i$ is a unit vector, $\mathbf{x}_i \cdot \mathbf{x}_i = 1$.



The position vector of Y_1

From $\overrightarrow{GY_1} = -\lambda\overrightarrow{GX_1}$ we get $\mathbf{y}_1 - \mathbf{g} = -\lambda(\mathbf{x}_1 - \mathbf{g})$, so

$$\begin{aligned}\overrightarrow{OY_1} = \mathbf{y}_1 &= (1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1 \\ &= \tfrac{1}{3}(1+\lambda)(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3) - \lambda\mathbf{x}_1 \\ &= \tfrac{1}{3}((1-2\lambda)\mathbf{x}_1 + (1+\lambda)(\mathbf{x}_2 + \mathbf{x}_3)),\end{aligned}$$

since $\frac{1}{3}(1+\lambda) - \lambda = \frac{1}{3}(1-2\lambda)$. This is the required expression.

Finding λ_1

Because Y_1 is on the circle, $|\mathbf{y}_1|^2 = 1$, so $|3\mathbf{y}_1|^2 = 9$. We need

$$|\mathbf{x}_2 + \mathbf{x}_3|^2 = 1 + 2\alpha + 1 = 2 + 2\alpha, \qquad \mathbf{x}_1 \cdot (\mathbf{x}_2 + \mathbf{x}_3) = \beta + \gamma.$$

Expanding $|3\mathbf{y}_1|^2$ as a dot product with itself therefore gives

$$(1-2\lambda)^2 + (1+\lambda)^2(2+2\alpha) + 2(1-2\lambda)(1+\lambda)(\beta+\gamma) = 9.$$

Collecting powers of λ and dividing by 2:

$$(3 + \alpha - 2\beta - 2\gamma)\lambda^2 + (2\alpha - \beta - \gamma)\lambda - (3 - \alpha - \beta - \gamma) = 0.$$

To keep this tidy, write

$$A = 3 + \alpha - 2\beta - 2\gamma, \qquad B = 3 - \alpha - \beta - \gamma,$$

and notice that the middle coefficient is exactly $A - B$. So the equation is $A\lambda^2 + (A - B)\lambda - B = 0$, which factorises:

$$(\lambda + 1)(A\lambda - B) = 0.$$

? **Ask yourself** Why should $\lambda = -1$ be a root, and why can we discard it?

Putting $\lambda = -1$ into $\mathbf{y}_1 = (1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1$ gives $\mathbf{y}_1 = \mathbf{x}_1$. Every point of the line through X_1 and G has the form $(1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1$, and this line meets the circle at X_1 as well as at Y_1, so the quadratic was always going to have $\lambda = -1$ as a root. We are told $\lambda_1 > 0$, so $\lambda_1 \neq -1$ and hence $A\lambda_1 = B$.

It remains to check we may divide by A. If $A = 0$ then $A\lambda_1 = B$ forces $B = 0$. But

$$|\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3|^2 = 3 + 2(\alpha + \beta + \gamma), \qquad \text{so} \qquad B = \tfrac{1}{2}(9 - |\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3|^2) = \tfrac{9}{2}(1 - |\mathbf{g}|^2).$$

By the triangle inequality $|\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3| \leq 3$, with equality only if the three unit vectors are equal; the points are distinct, so $B > 0$. Hence $A \neq 0$ and

$$\lambda_1 = \frac{B}{A} = \frac{3 - \alpha - \beta - \gamma}{3 + \alpha - 2\beta - 2\gamma},$$

as required.

The final part

Since $\overrightarrow{GY_1} = -\lambda_1 \overrightarrow{GX_1}$ with $\lambda_1 > 0$, the lengths satisfy $GY_1 = \lambda_1 GX_1$, so

$$\frac{GX_1}{GY_1} = \frac{1}{\lambda_1} = \frac{3 + \alpha - 2\beta - 2\gamma}{3 - \alpha - \beta - \gamma}.$$

The same argument with the labels cycled ($1 \to 2 \to 3 \to 1$, which sends $\alpha \to \beta \to \gamma \to \alpha$) gives the other two ratios. The denominator $B = 3 - \alpha - \beta - \gamma$ is symmetric, so it is the same for all three, while the numerators are

$$3 + \alpha - 2\beta - 2\gamma, \qquad 3 + \beta - 2\gamma - 2\alpha, \qquad 3 + \gamma - 2\alpha - 2\beta.$$

These add up to $9 + (1 - 2 - 2)(\alpha + \beta + \gamma) = 9 - 3(\alpha + \beta + \gamma) = 3B$. Therefore

$$\frac{GX_1}{GY_1} + \frac{GX_2}{GY_2} + \frac{GX_3}{GY_3} = \frac{3B}{B} = 3.$$

Checks

- Equilateral triangle: then $G = O$, each chord through G is a diameter, and every ratio is 1. In the formula, $\alpha = \beta = \gamma = -\frac{1}{2}$, giving $\lambda_1 = \frac{9/2}{9/2} = 1$.
- There is a geometric reading: $A = \frac{1}{2}|\mathbf{x}_2 + \mathbf{x}_3 - 2\mathbf{x}_1|^2 = \frac{9}{2} GX_1^2$ and $B = \frac{9}{2}(1 - OG^2)$, so $\lambda_1 = B/A$ says $GX_1 \cdot GY_1 = 1 - OG^2$, the intersecting chords (power of a point) theorem.
- Numerically: with X_1, X_2, X_3 at angles 0, 1 and 2.5 radians, the three ratios GX_i/GY_i come out as about 1.13, 0.31 and 1.57, and their sum is 3 to machine precision. We tested 20 000 random triangles the same way.

Common slips

- Forgetting that $|\mathbf{x}_2 + \mathbf{x}_3|^2$ contains the cross term 2α, not just $1 + 1$.
- Grinding through the quadratic formula instead of spotting the root $\lambda = -1$.
- Dividing by $3 + \alpha - 2\beta - 2\gamma$ without saying why it is non-zero.

Beyond STEP

The identity $GX_1 \cdot GY_1 = 1 - OG^2$ hiding in this question is the *power of a point* with respect to a circle, and the dot-product calculation is the vector proof of it. Writing geometry in terms of inner products, where lengths and angles all come from $\mathbf{u} \cdot \mathbf{v}$, is the starting point for inner-product spaces in a first linear algebra course. Strang's *Introduction to Linear Algebra* develops this viewpoint.

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